A Profiled Heading Candidate for Single-Tag Range–Inertial Position Initialization

Abstract— **A single ultra-wideband tag measures distances to surveyed anchors at different times. Given gravity alignment and nominal gyro bias, we use the intervening inertial motion to initialize position at the latest packet with unknown heading. Eliminating polynomial translation and auxiliary terms from a squared-range relaxation leaves a unit-circle problem. Its solution supplies one heading candidate for refinement on the original distances. We give a sufficient condition for noiseless recovery in the relaxation and derive how rotation separates a body-fixed accelerometer offset from a free quadratic correction. On 66 flight windows evaluated under 528 correlated configurations, one candidate and four uniform heading starts attain the same best tested objective within tolerance and the same horizontal root-mean-square error of 0.159 m. A local start misses that objective in seven configurations and gives 0.154 m. On 18 timed windows, median serial solve times including candidate generation are 30.7 ms for the proposed candidate, 30.6 ms for a local start and 135.4 ms for four starts. In a separate cold-start replay, the published continuous-time spline estimator SFUISE confirms a usable horizontal position earlier on both datasets.**

Index Terms— **Inertial navigation, invariant extended Kalman filter, observability, position acquisition, ultra-wideband ranging.**

I. Introduction

A quadrotor emerging from radio shadow receives one range at a time. Each packet constrains the antenna to a sphere at the instant of the exchange. By the time another anchor replies, the aircraft has moved. Treating these ranges as a set of simultaneous spheres introduces a motion error. Integrating the inertial measurement unit (IMU) between exchanges links the antenna positions, but an error in tilt or accelerometer bias bends the integrated path. The bend grows with the age of the oldest packet. A longer window can therefore improve the geometry of multilateration while making the assumed inertial trajectory less accurate.

Published range-only methods recover a trajectory by imposing a basis on its positions [1]. A model of quadrotor thrust and drag supplies motion information to a moving-horizon tracker [2]. Continuous-time spline and factor-graph estimators can initialize from single-tag radio and inertial measurements [3], [4]. We study how to choose a heading seed when estimating endpoint position from a short window of such data. The model assumes surveyed anchors, reliable gravity alignment and a usable nominal gyro bias. Initial position, absolute yaw and endpoint velocity are unknown. An IMU prefix provides a practical way to estimate tilt and bias when its motion permits. Writing the resulting fix into a running navigation filter poses a further question because velocity and attitude also determine the next predicted position.

Our formulation uses two pieces of information available to a single tag. The ranges locate the aircraft relative to the anchors, and the gravity-aligned IMU records the shape of its motion up to an unknown heading. In a squared-range relaxation, we eliminate the endpoint translation, velocity and auxiliary products analytically. The remaining unit-circle problem supplies a heading candidate for a single nonlinear fit to the original distances. Its global solution applies to the relaxed candidate objective.

The second contribution concerns the motion that remains informative after the translation and bias are fitted. We show when a constant body-fixed accelerometer offset and a free world-frame quadratic produce the same displacement correction, how their difference grows with rotation, and which local yaw variation the nuisance states can absorb. We compare the candidate with ordinary local and uniform multistart optimization on the same objective. Separate experiments compare first usable position with published estimators and illustrate how a checkpointed invariant extended Kalman filter (InEKF) uses a window after a range outage.

II. Related work

A. Trajectory reconstruction and observability

Batista *et al.* obtained observability conditions and a filter for single-range navigation with measured relative velocity [5]. Theodosis *et al.* showed how quadratic outputs of certain linear systems admit a finite-dimensional linear immersion [6]. Their single-anchor navigation example takes world acceleration as an input. Both results explain how motion supplies information missing from an instantaneous range. On a quadrotor, the accelerometer reports body-frame specific force. Converting it to world acceleration already requires the heading and bias that the window is asked to estimate.

Pacholska *et al.* used an assumed trajectory basis to recover continuous position from distances alone [1]. Lifting products of basis coefficients makes the algebra linear and leads to explicit recovery conditions. Goudar *et al.* developed range-only trajectory initializers with guarantees under specified configurations of tags, beacons and motion [7]. A known baseline between body-mounted radios can supply attitude information that a single colocated tag lacks. Here, the known component is the trajectory shape obtained by integrating an IMU in a gravity-aligned frame. Separating the coefficients that enter linearly follows the variable-projection principle of Golub and Pereyra [8]. Our use of that principle leaves two coupled coefficients on a unit circle. The physically admissible raw-range trajectories form a smaller set than those admitted by the squared-range relaxation.

B. Asynchronous fusion and inertial bias

Tightly coupled UWB/IMU estimators directly relate the radio measurement to a moving pose. Hol *et al.* connected heading estimation to acceleration excitation [9]. Kok *et al.* developed an optimization-based estimator with an asymmetric time-of-arrival error model [10]. Li *et al.* fitted continuous-time splines to asynchronous UWB and IMU samples [3]. Jiang *et al.* initialized a joint range–inertial graph and incrementally optimized it on recorded single-sided two-way ranging (TWR) data [4]. Incremental smoothing can efficiently relinearize a trajectory as new factors arrive [11]. These methods represent motion throughout a trajectory. Our window compresses the translational uncertainty into endpoint position, endpoint velocity and three acceleration-correction coefficients. This makes it possible to compare two physical explanations for the same range history at equal model dimension.

IMU preintegration incorporates the sensitivity of integrated increments to sensor biases [12]. We use that sensitivity in an endpoint-centered form. It has two consequences for acquisition. The leading position perturbation can agree with a constant world acceleration even when the true offset is body-fixed. With strong rotation, an unrestricted world acceleration and a constant body bias produce different families of positions. The heading question is consequently about which component of a change in yaw survives after fitting all the motion coefficients.

C. Radio calibration and navigation benchmarks

Radio error also varies systematically with geometry and hardware. Prorok and Martinoli learned spatial error models for indoor ranging [13]. Ledergerber and D'Andrea used maximum likelihood to calibrate systematic ranging errors [14]. Shalaby *et al.* characterized antenna delays and power-dependent bias and variance in TWR [15]. Zhao *et al.* combined learned bias correction with robust filtering for time differences of arrival (TDoA) [16]. Goudar and Schoellig studied excitation conditions for estimating the spatial and temporal offsets between a radio and an IMU [17]. Ferrero-Guillén *et al.* analyzed a further source of error arising from motion during the messages of an asynchronous TWR exchange [18]. Evaluating a finished range at its transaction time addresses motion *between* packets. It leaves motion within the exchange and unmeasured antenna delays in the observation error. We therefore compare motion models under a common radio calibration and examine transfer with its coefficients removed.

Pfeiffer *et al.* released a vehicle-model moving horizon estimator (MHE), a companion extended Kalman filter (EKF) and a quadrotor dataset [2], [19]. Their later MHE uses adaptive step sizes and switching variables for outlier handling [20]. We test the released 2021 classes at matched endpoint times and retain their dynamics and attitude handling. Oursland and Mehrabian [21] used an InEKF for a tag with a known lever arm and signed TDoA. Invariant error coordinates can improve the linearization of navigation dynamics under the hypotheses studied by Barrau and Bonnabel [22]. They cannot supply an independent heading measurement. We use the native TDoA filter on recordings with measured attitude, and assess the zero-lever-arm TWR acquisition model on separate flights.

Table I relates each method's motion input to its output. We compare position at the last packet available to every estimator.

III. A yaw candidate from asynchronous ranges

A. Endpoint coordinates

Let R map body vectors to a navigation frame with vertical e_3, and write v, p for the velocity and position of the IMU. The accelerometer and gyro report f_m, ω_m with biases b_a, b_g. For nominal propagation,

$$\dot{R} = R[\omega_m - b_g]_\times, \quad \dot{v} = R(f_m - b_a) + g, \quad \dot{p} = v, \tag{1}$$

where $g = -9.81 e_3$ in m/s^2 and $[x]_\times y = x \times y$. A TWR transaction i supplies range z_i to surveyed anchor a_i at time t_i. The window takes the antenna and IMU to be colocated. With $D_i = \|p_i - a_i\|$ and $n_i = (p_i - a_i)/D_i$, its observation model is

$$z_i = D_i + \mu_{a_i}(D_i, n_i) + e_i. \tag{2}$$

Here the anchor subscript on μ identifies a radio link. The scale σ_i describes the inlier range error. The fit also accommodates occasional large residuals.

Write $R(t) = R_z(\psi)Q(t)$, where Q follows nominal gravity alignment and gyro integration in an arbitrary initial yaw frame. Let $s(t)$ be a nominal double integral of acceleration in that frame and set T to the last available sample. Defining $\tau_i = t_i - T$ gives

$$\begin{aligned} d_i &= s(t_i) - s(T) - \tau_i \dot{s}(T), \\ p_i &= p_T + \tau_i v_T + R_z(\psi) d_i. \end{aligned} \tag{3}$$

Reanchoring removes the integration constants even when the starting velocity is unknown. Both $d(T)$ and $\dot{d}(T)$ vanish. Gravity commutes with the yaw rotation, so d_i includes its displacement. The design evaluates the held IMU model at each transaction time. No simultaneous group of anchor readings is required. We call (3) the *affine* model because its free translation is affine in time. Its nominal inertial path can be highly curved.

B. Lifting and elimination of translation

For the seed, write $p_i = c(\tau_i) + R_z(\psi)d_i$, where $c(\tau) = \sum_{j=0}^{K} c_j \tau^j$. Acquisition uses $K = 1$. The general degree clarifies what information a more flexible translation removes. Subtract a constant radio offset to obtain $r_i = z_i - \beta_{a_i}$ (zero offset when uncalibrated), and form

$$y_i = \tfrac{1}{2}(r_i^2 - \sigma_i^2 - \|a_i\|^2 - \|d_i\|^2) + a_{iz} d_{iz}. \tag{4}$$

In the noiseless identity take $\sigma_i = 0$. For $u = (\cos\psi, \sin\psi)^\mathsf{T}$, expansion then yields

$$y = B\nu + Dc + Cu. \tag{5}$$

TABLE I

Motion assumptions and estimation targets in related range-based navigation methods.

Approach	Motion information	Estimation target
Range-only lifting [1]	Assumed finite trajectory basis	Continuous single-tag position
Range-only initialization [7]	Rigid tag geometry and a prescribed motion model	Pose or trajectory under tag and beacon geometry conditions
UWB–inertial smoothing [3], [4]	IMU samples with a spline or factor graph	Trajectory and inertial states over the window
Quadrotor MHE and EKF [2], [20]	Vehicle-specific thrust and drag models	Rolling or recursive position and velocity
TDoA InEKF [21]	IMU propagation and an antenna lever arm	Recursive attitude, velocity and position
Single-tag acquisition (this work)	Measured inertial shape and a rotating body-fixed offset	Endpoint position and velocity with unknown initial heading

The vector c stacks the c_j. The row of B contains τ_i^ℓ for $0 \le \ell \le 2K$, followed by the pairs $(\tau_i^j d_{ix}, \tau_i^j d_{iy})$ for $0 \le j \le K$. The remaining rows are

$$\begin{aligned} D_{ij} &= \tau_i^j[-a_{ix}, -a_{iy}, d_{iz} - a_{iz}], \\ C_i &= [-a_{ix}d_{ix} - a_{iy}d_{iy},\ a_{ix}d_{iy} - a_{iy}d_{ix}]. \end{aligned} \tag{6}$$

For a physical path, the coefficients in ν are constrained. The coefficient of τ^ℓ in $\|c(\tau)\|^2/2$ is $\frac{1}{2}\sum_{j+k=\ell} c_j^\mathsf{T} c_k$. The two horizontal cross-term coefficients for each j are the components of $(R_z(\psi)^\mathsf{T} c_j)_{xy}$. Releasing these identities makes ν independent of c and u. This is the only relaxation in (5).

For an ideal zero-mean Gaussian inlier, the centered half-squared range has variance $D_i^2\sigma_i^2 + \sigma_i^4/2$. We substitute $\max(r_i, \sigma_i)$ for the unknown distance in this expression and place the inverse standard deviations on the diagonal of W. With $L = W[B\ D]$, define $P = I - LL^\dagger$, $\bar{y} = PWy$ and $\bar{C} = PWC$. Minimizing over all the linear coefficients gives

$$F(\psi) = \left\|\bar{y} - \bar{C}u(\psi)\right\|^2. \tag{7}$$

This projection is exact within the relaxed squared-range model at fixed weights. The implementation removes the left singular vectors whose singular values exceed 10^{-10} times the largest value. It never needs to materialize the dense projector.

Set $H = \bar{C}^\mathsf{T}\bar{C},\ h = \bar{C}^\mathsf{T}\bar{y}$ and $\Delta = H_{22} - H_{11}$. The terms varying with yaw are

$$\begin{aligned} F(\psi) = &-\tfrac{\Delta}{2}\cos 2\psi + H_{12}\sin 2\psi \\ &- 2h_1\cos\psi - 2h_2\sin\psi + \text{const}. \end{aligned} \tag{8}$$

Only the first two angular harmonics remain. With $q = \tan(\psi/2)$, setting the derivative to zero gives

$$\begin{aligned} 0 = &(H_{12} + h_2)q^4 + 2(h_1 - \Delta)q^3 - 6H_{12}q^2 \\ &+ 2(h_1 + \Delta)q + (H_{12} - h_2). \end{aligned} \tag{9}$$

Checking the real roots and $\psi = \pi$ covers the circle. The implementation also checks zero for a degenerate polynomial. We call the policy that retains one minimum of (7) *Profile-1*. Its global solution concerns the seed objective. The subsequent robust distance fit can have distinct local minima.

C. Identifiability and sensitivity of the heading seed

For an exact noiseless model the physical $u_\star$ makes (7) zero. If $\bar{C}$ has column rank two, every minimizer obeys $\bar{C}(u - u_\star) = 0$ and hence equals $u_\star$. Once yaw is known, translation in the lifted equations is unique if $(I - WB(WB)^\dagger)WD$ has rank $3(K+1)$. These are sufficient conditions. A rank-one projected system may intersect the circle twice, or tangentially at one point.

The row count gives a practical interpretation. For $K = 1$ the lift has seven auxiliary and six translation columns. If these 13 columns are independent, at least 15 transactions are needed to leave two projected directions. Adding a quadratic raises these counts to 11 and nine, hence 22 transactions in general position. Collecting that many packets does not guarantee useful geometry. Poor anchor height diversity or a nearly polynomial inertial shape can leave the smallest singular value of $\bar{C}$ very small. Even perfect arithmetic on (9) then gives little protection against an error in d_i.

There is also a finite perturbation bound. At fixed design let $\bar{y} = \bar{C}u_\star + \epsilon$, and let $\hat{u}$ minimize (7). Feasibility of $u_\star$ gives $\|\bar{y} - \bar{C}\hat{u}\| \le \|\epsilon\|$. The triangle inequality and the chord length on the unit circle then yield

$$\left|\sin\frac{\hat{\psi} - \psi_\star}{2}\right| \le \frac{\|\epsilon\|}{\sigma_{\min}(\bar{C})} \quad \text{if } \operatorname{rank}(\bar{C}) = 2. \tag{10}$$

The denominator can be computed before any nonlinear fit. The numerator is the departure from the physical heading in the projected equations, which can include inertial mismatch. This relation explains why concentrating the anchors in a shallow plane and adding more starts address different sources of error.

IV. Fitting the physical trajectory

A. A rotating accelerometer offset

Suppose the true accelerometer bias differs from the nominal value by a constant δb over the window. At fixed nominal rotation its contribution to world acceleration is $-R_z(\psi)Q(t)\delta b$. Integrating twice and subtracting the endpoint position and velocity contributions gives

$$B_i^a = \int_{t_i}^{T} (s - t_i)Q(s)\,ds, \tag{11}$$

$$p_i = p_T + \tau_i v_T + R_z(\psi)(d_i - B_i^a \delta b). \tag{12}$$

The sign of the kernel follows from integrating backward from T with zero terminal displacement and velocity. The integration in (11) is exact for the held rotation samples in the estimator. A changing gyro bias or an incorrect tilt changes the nominal rotation itself and can produce errors outside this three-parameter family. We interpret the fitted coefficient as a short-window motion correction.

On flights with independent radio calibration data we use the mean

$$\mu_a(D, n) = \beta_a^g + s_a(D - \bar{D}_a) + \kappa_a^{\mathsf{T}} n. \tag{13}$$

The link intercept, slope and direction coefficients are robust ridge estimates from training flights. The simpler constant-offset model uses a training median β_a. Both use the same training median-absolute-deviation range scales. A line-of-sight coefficient can absorb an apparent anchor displacement, since a small displacement δa changes distance by $-n^{\mathsf{T}}\delta a$ to first order. Its presence in (13) therefore does not identify a physical survey error. We freeze these coefficients before scoring and remove the whole mean at a site without calibration.

Define $r_i^p = (D_i + \mu_{a_i} - z_i)/\sigma_i$, and let $\rho_2(r)$ equal $r^2/2$ for $|r| \le 2$ and $2|r| - 2$ otherwise. The endpoint fit minimizes

$$\min_{p_T, v_T, \psi, \delta b} \sum_i \rho_2(r_i^p) + \frac{\|\delta b\|^2}{2\sigma_b^2}, \qquad \sigma_b = 0.10 \text{ m/s}^2. \tag{14}$$

The prior on the three bias components stays quadratic. We first fit a constant-velocity translation to the ranges from two starts, a squared-range linear solution and a stationary anchor-centroid start. Only the better translation enters (14). Profile-1 supplies its yaw and δb starts at zero. Thus the principal acquisition method performs two common translation fits and one final ten-variable distance fit. The raw fit has a cap of 120 function evaluations per start. The affine control follows the same seed policy with $\delta b = 0$.

Differentiating the radio mean matters when the aircraft crosses a learned direction gradient. Write $\ell_i = R_z(\psi)(d_i - B_i^a \delta b)$. The two factors of the residual Jacobian are

$$\begin{aligned} g_i &= (1 + s_a) n_i + (I - n_i n_i^{\mathsf{T}}) \kappa_a / D_i, \\ G_i &= [\, I,\ \tau_i I,\ e_3 \times \ell_i,\ -R_z(\psi) B_i^a \,], \\ J_i &= \sigma_i^{-1} g_i^{\mathsf{T}} G_i. \end{aligned} \tag{15}$$

For the constant radio mean $g_i = n_i$. The second term of g_i acts perpendicular to the line of sight and accounts for the change in predicted direction when the fitted position moves.

B. Local equivalence to a quadratic correction

Let $h_i = T - t_i$. A freely fitted world acceleration c_W adds $\frac{1}{2} h_i^2 c_W$ to (3). It costs three parameters, the same number as δb, and has no prior. For a fixed attitude the two motion models span exactly the same corrections through $c_W = -R_z(\psi) Q(T) \delta b$. When $\|\dot{Q}\|_2 \le \Omega$, the difference between their leading terms obeys

$$\left\| B_i^a - \tfrac{1}{2} h_i^2 Q(T) \right\|_2 \le \int_0^{h_i} (h_i - u) \Omega u \, du = \tfrac{1}{6} \Omega h_i^3. \tag{16}$$

If Q is twice continuously differentiable, expanding (11) gives $B_i^a = \frac{1}{2} h_i^2 Q(T) - \frac{1}{6} h_i^3 \dot{Q}(T) + O(h_i^4)$. Rotation first changes the shape at cubic order. In Fig. 1, the two displacement corrections differ by 0.078 m at 2 s and 0.575 m at 4 s. A short or weakly turning segment can benefit from correcting acceleration before it distinguishes the two models.

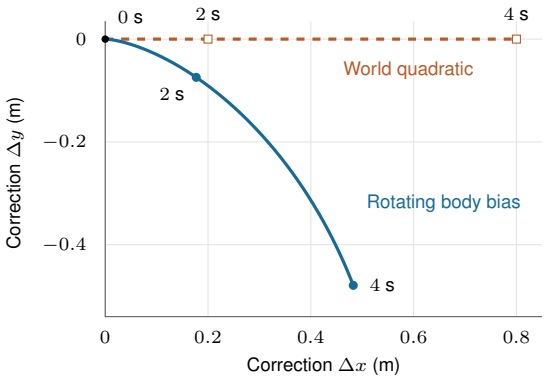


Fig. 1. Displacement corrections in an analytic planar turn. Both start with the same terminal correction acceleration $+0.10 e_x$ m/s^2. At a yaw rate of 0.60 rad/s the body-fixed correction curves away from the world-frame quadratic. Markers give time before the endpoint. Both axes have the same spatial scale.

C. Heading sensitivity after fitting nuisance states

For local analysis, multiply each radio Jacobian row in (15) by $\sqrt{\min(1, 2/|r_i^p|)}$, taking weight one at zero. Partition the result into the yaw column $\tilde{j}_\psi$ and the position, velocity and correction columns $\tilde{J}_n$. To first order, the part of a yaw change that these nuisance variables cannot absorb has squared norm

$$\mathcal{I}_{\psi|n} = \min_a \left\| \tilde{j}_\psi - \tilde{J}_n a \right\|^2 = \left\| (I - \tilde{J}_n \tilde{J}_n^{\dagger}) \tilde{j}_\psi \right\|^2. \tag{17}$$

This is a local Gauss–Newton curvature with fixed robust weights. Adding the bias prior penalizes the minimizing nuisance motion and can increase this number without adding information from the ranges. A locally sharp minimum can also coexist with a distant yaw mode.

A limiting case exhibits exact ambiguity. Assume constant Q, constant measured specific force f_m, zero lever arm and a freely fitted constant bias. For any second yaw ψ', choose

$$b_a' = f_m - Q^{\mathsf{T}} R_z(\psi')^{\mathsf{T}} R_z(\psi) Q (f_m - b_a). \tag{18}$$

Both pairs (ψ, b_a) and (ψ', b_a') generate the same world acceleration. With the same starting position and velocity, their positions, radio directions and all ranges coincide even for ideal anchors. A bias prior can select between them. The radio data cannot. A time-varying f_m generally

makes the right side of (18) time dependent, so constant attitude alone is insufficient for this ambiguity.

A related loss occurs with free polynomial translation. Adding a vector polynomial $b(\tau)$ of degree at most K to d and replacing $c(\tau)$ by $c(\tau) - R_z(\psi)b(\tau)$ preserves every position for each fixed yaw. If d lies entirely in this polynomial space, varying yaw cannot change the best range fit. The bias model in (14) restricts its allowed correction through B_i^a and a prior.

V. Flight data and comparison protocol

A. Acquisition windows and reference positions

The Delft dataset [19] contains two TWR recordings for each of six motions. We use the earlier recording of each motion to fit radio calibration and the later one to score acquisition. All six evaluation flights share the room and campaign date and were available during method development. A held-motion calibration control excludes the earlier recording of the target motion as well. It tests whether a directional calibration generalizes to a different motion in the same anchor installation.

The replay clock starts after a 1.5 s IMU prefix for tilt and nominal bias. An offline 0.20 s offset aligns OptiTrack for scoring at the actual last sample of each window. Reference positions select the supported interval and are withheld from the estimator. We test 2 and 4 s windows ending at 5, 10, 20 and 30 s, adding 8 s windows at the last three times. The six flights yield 66 physical windows and 24 distinct endpoint times. Testing all eight anchors and the four lowest gives 132 layout-specific windows.

For the seed comparison we also solve in four equivalent inertial yaw coordinate frames. The same physical measurements appear in each frame, although a finite-budget optimizer can reach different minima. The 528 configurations characterize solver behavior across coordinates and layouts. We additionally fix one coordinate frame for flight comparisons. The cold-start test in Section D uses a single native start per recording.

The outdoor QDrone dataset [23] contributes ten recordings from a field, a building, a bridge and a tunnel. Nested 2, 4 and 8 s windows end at the midpoint of sensor coverage in each recording, giving 30 segments at ten endpoint instants. The published total-station prism position serves as a proxy because its exact transform to the antenna is unavailable. Both fits use the same IMU prefix, zero radio mean and a 0.30 m radio scale. We use the registered coordinates directly, without site-specific radio fitting or truth-based realignment.

B. Metrics and matched inputs

For window i, let $e_{i,xy} = \left\|\hat{p}_{i,xy} - p^*_{i,xy}\right\|$. For an explicitly specified set S we report

$$E_{xy}(S) = \sqrt{|S|^{-1} \sum_{i \in S} e^2_{i,xy}}. \tag{19}$$

TABLE II

Endpoint RMSE in metres with one profiled yaw candidate per fit. Delft pools all four yaw coordinates. QDrone uses the prism proxy.

	Affine			Body bias		
Population	XY	Z	3D	XY	Z	3D
Delft, eight	1.585	.758	1.757	.116	.209	.239
Delft, low four	1.664	1.163	2.030	.192	.526	.560
Delft, pooled	1.625	.981	1.898	.159	.400	.431
QDrone	2.930	1.655	3.365	1.631	1.779	2.414

Vertical (Z) and three-dimensional (3D) errors use the corresponding components of the position residual. Within the same set $E^2_{3\mathrm{D}} = E^2_{xy} + E^2_z$, which allows a vertical failure to be distinguished from a horizontal one. We use flight-level pairs to show heterogeneity and retain the larger set of fits where measuring optimization reliability is the purpose. The reported summaries are descriptive. Overlapping windows are correlated, as are alternative layouts of the same anchor field.

For the Delft fixed-window endpoint comparisons, the estimators restart at the same window boundary and receive the same positive range updates and anchor coordinates. We supply either raw ranges with a 0.25 m scale or training-corrected constant offsets and scales. The released MHE uses 20 internal nodes at 50 Hz, one gradient step and sample-consensus rejection. Its gradient ignores the supplied variances. The released EKF runs at 100 Hz. Both retain the authors' zero translation, level zero-yaw attitude and thrust/drag model with an acceleration-based thrust proxy. We disable auxiliary height and consume pending final packets. Our method uses the IMU prefix for tilt and searches yaw.

For continuous tracking we compute a time-domain XY RMSE from the most recently emitted state on a 20 Hz grid within each flight and interval. Section VII reports medians over reference-supported flights and paired differences between policies.

VI. Endpoint accuracy and competing estimators

A. Effects of trajectory correction and radio calibration

Table II compares the affine and bias-corrected fits under the same single-candidate policy. This Delft ablation changes both the motion correction and the radio mean. In QDrone the mean is zero for both fits. Table III separates their contributions under common calibration.

Across Delft configurations, horizontal RMSE falls by 90.2%. The correction improves 440 of 528 paired errors, including all 144 configurations with an 8 s window. At 2, 4 and 8 s, pooled XY changes from 0.222, 0.709, 2.991 m to 0.164, 0.175, 0.126 m. With the yaw coordinate fixed, all six flights improve (Fig. 2) and pooled XY falls from 1.645 to 0.159 m.

On QDrone, motion correction reduces pooled horizontal error to the registered prism by 44.3% without

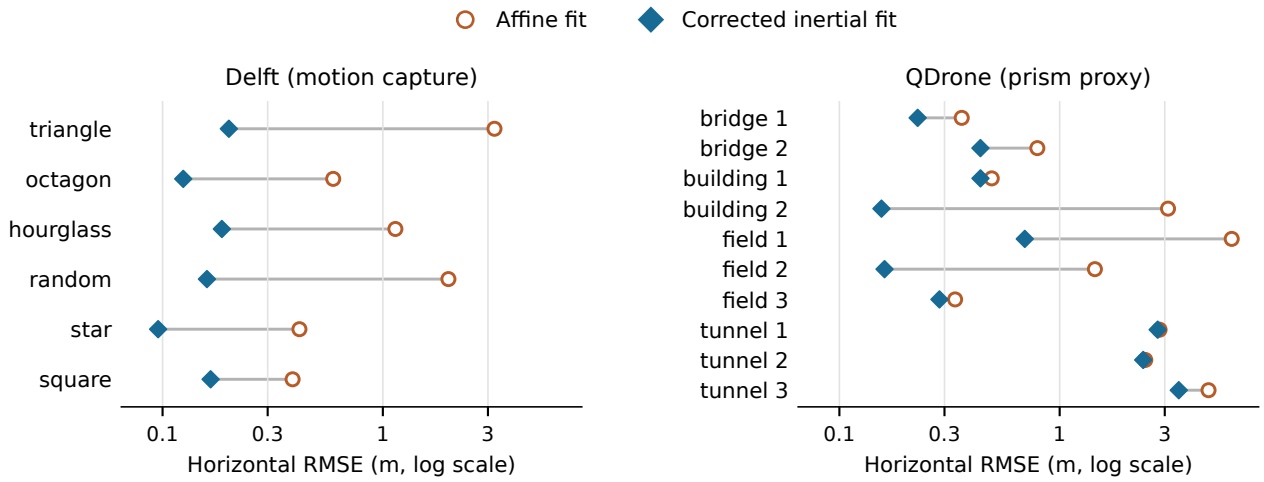


Fig. 2. Horizontal endpoint RMSE by recording. Each line joins an affine fit to a corrected inertial fit at the same endpoints. Delft includes 22 errors per flight at one yaw coordinate and adds a trained radio mean to the correction. QDrone pools three windows per recording and changes only the motion model. The horizontal axis is logarithmic.

TABLE III

Delft control for motion and radio calibration. Four uniform yaw starts, reference coordinate and 66 physical windows per layout. Entries are RMSE in metres.

		Eight anchors		Low four	
Motion	Radio	XY	3D	XY	3D
Affine	Constant	1.631	1.799	1.665	2.029
Quadratic	Constant	.127	.320	.201	.924
Body bias	Constant	.133	.279	.203	.874
Affine	Directional	1.654	1.756	1.653	1.924
Quadratic	Directional	.111	.273	.188	.591
Body bias	Directional	.116	.239	.192	.560

TABLE IV

Delft seed policies on the same 528 configurations and body-bias objective. Evaluations include both common translation fits. Capped final solutions are retained.

Yaw policy	Misses	Evaluations	XY (m)	3D (m)
One local yaw	7	36,313	.154	.429
Profile-1	0	32,789	.159	.431
Four uniform yaws	0	86,508	.159	.431

site calibration. The 2, 4 and 8 s values change from 1.634, 2.120, 4.312 m to 1.659, 1.629, 1.605 m. Pooling the three lengths gives an improvement in every recording.

With four starts held fixed (Table III), the affine fit exceeds 1.6 m in XY under both radio models. Either three-parameter motion correction removes most of that error. Under directional calibration the world quadratic and body bias give 0.155 and 0.159 m in pooled XY, winning three flights each. Their 3D errors are 0.460 and 0.431 m. These results agree with the leading-order equivalence in (16). On QDrone the corresponding XY errors are 1.677 and 1.631 m.

Calibration adds a benefit that survives holding out the target motion from training. Changing the body-model mean from constant to directional changes XY from 0.135 to 0.119 m with eight anchors and from 0.204 to 0.199 m with the four lowest. The corresponding 3D changes are 0.289 to 0.250 m and 0.864 to 0.616 m. The larger 3D gain with low anchors reflects the sensitivity of height to the range mean.

B. Cost and reliability of the yaw candidate

We next fix the body model and compare its observed objective for a local yaw, Profile-1 and four uniform yaw starts. Let J denote the stored objective in twice the normalization of (14). A miss on a real window means $J - J_{\min} > 10^{-4}(1 + J_{\min})$, where $J_{\min}$ is the best among these three policies for that same window. Raw objectives across different motion families have different meanings and are never used for this count.

Profile-1 reaches the best observed objective within tolerance everywhere in Table IV, using 62.1% fewer nonlinear function evaluations than four starts. A single local yaw misses seven times and gives a slightly lower pooled horizontal error. These counts include the two common translation fits. Constructing the projected candidate requires additional linear algebra outside the evaluation count. On the outdoor recordings the miss counts for local, profiled and four-start fits are 1, 0 and 0.

We timed all four policies on the six Delft flights using the same endpoint at 10 s, 2/4/8 s windows, all eight anchors, the reference yaw coordinates and fixed directional calibration. Each invocation takes the common record-level inertial integrals as input, extracts the window, computes its bias sensitivity, executes both translation fits, constructs the candidates and performs all distance

refinements. Prefix alignment and common record-level integration precede the timer. In one process with single-threaded BLAS, we mixed the policy order over three warm serial rounds. Taking the median on each window and then across 18 windows gives 30.6, 30.7, 76.3 and 135.4 ms for a local start, Profile-1, two starts and four starts, respectively. On this subset all four policies reach the best tested objective within tolerance and give 0.104 m horizontal RMSE. The larger experiment in Table IV includes the seven local-start misses.

A simulation tests the affine-model seed 30 m horizontally from the centroid of eight anchors at $(\pm 4, \pm 4, z)$ m, $z \in \{0, h\}$. Each of 200 windows per geometry contains 4 s of inertial motion and 240 ranges with 0.10 m Gaussian noise. Success is within $\max(10^{-3}, 10^{-5}|J_{\min}|)$ of the best of five policies (local, Profile-1 and 2, 4 or 12 uniform starts). At $h = 3$ m, Profile-1 and two starts both have zero misses and 0.509 m 95th-percentile 3D error. Their median times are 47.1 and 66.6 ms in 40 serial trials per policy on an i5-12600KF with single-threaded BLAS. At $h = 0.1$ m they miss 12 and 10 times, respectively. Shallow geometry also raises the Profile-1 95th-percentile 3D error to 3.946 m.

C. Range-only and vehicle-model estimators

The relaxation of Pacholska *et al.* [1] reconstructs motion using only the ranges. We evaluate the published line basis $(1, t)$ and a prespecified quadratic $(1, t, t^2)$ in the same solver. The latter represents the entire path, whereas the quadratic in Table III corrects a measured inertial path. Under common constant calibration, incorporating the IMU and fitting the body offset reduces horizontal RMSE from 0.356 to 0.133 m with eight anchors and from 0.474 to 0.204 m with the four lowest (Table V). On the low layout the corresponding vertical errors are 7.982 and 0.854 m. The range-only linear system has full numerical rank in every window. The inertial shape supplies additional structure when shallow sight lines carry little independent height information.

Fig. 3 shows where the improvement occurs. On the eight-anchor layout, the horizontal error of the range-only quadratic rises from 0.142 m at 2 s to 0.591 m at 8 s. The inertial fit gives 0.121 and 0.113 m at those durations. With the four lowest anchors the corresponding 8 s errors are 0.685 and 0.179 m. The additional IMU input allows longer segments to retain measured turns, while a range-only polynomial must explain the entire maneuver using the same few coefficients.

With eight anchors and constant correction, the author EKF and MHE of Pfeiffer *et al.* [2] achieve 0.105 and 0.121 m horizontal RMSE, compared with 0.133 m for the proposed window. In 3D the respective errors are 0.254, 0.343 and 0.279 m, with 65, 56 and 63 of 66 endpoints within 0.5 m. The window uses no thrust, mass or drag parameters. Evaluation-capped fits are included.

D. Cold start with unknown position and heading

We replay the six Delft flights and two handheld ISAS absolute-range recordings from Li *et al.* [3]. ISAS Walk1 supplies the tracker-to-IMU and world-frame reference calibration for Walk2 and Walk3. Within each recording the methods receive the same IMU, available range packets, surveyed anchors, tag offset and radio calibration. Each starts without an externally supplied position, heading or height. All have access to the same 1.5 s IMU prefix. The Pfeiffer implementations propagate their zero seeds during this prefix. Delft has a position reference only, available 0.46–2.39 s after prefix release across flights. On ISAS the range prediction includes the measured tag lever arm, $R(t_i)\ell$, and the position score refers to the IMU origin.

On the sensor clock, a fix is confirmed at the third consecutive 0.5 s grid point with horizontal error at most 0.5 m, using the latest estimate available at each point. The reference is used only for scoring. Confirmation measures observation time from the first common IMU sample. We also score horizontal error 10 s after prefix release and count recordings that stay within the threshold at every scored grid point from confirmation through 15 s after release. Late reference coverage can delay the first observed confirmation.

SFUISE uses a port of the authors' spline, residuals and optimizer. The adapter replaces ROS transport and repeats the endpoint control point at the single-knot boundary. The Jiang row uses our independent implementation of the published 10 s initialization and cumulative batch refits thereafter. The authors use incremental D-iSAM2 for subsequent updates. The Pfeiffer rows use the authors' quadrotor estimators with an accelerometer-derived thrust input. These measurements describe the implementations replayed here.

The profiled window confirms a position 8 s earlier than the fixed-duration graph, in median observation time on both datasets (Table VI). SFUISE gives earlier median confirmation on both, and lower 10 s XY error on four of the six flights and both walks. Under the same candidate policy the earlier affine window gives 10 s errors of 0.440 and 3.161 m. The body-bias correction improves seven of eight paired recordings. Its 10 s median heading error on the two walks is nevertheless 73.8°. A position fix alone supplies insufficient evidence for a heading reset.

We also tested whether feeding an erroneous prior back to the window assists recovery. Each recording contributes one midpoint restart. Initial XY offsets of 1, 5, 10 and 20 m in four directions are crossed with heading offsets of 0°, 90° and 180°. Delft lacks measured attitude, so its heading offsets use the common inertial yaw frame. A Gaussian factor constrains the physical initial state of a 10 s window. With standard deviations of 20 m/120° and 0.10 m/5° respectively, the Delft window reaches 0.5 m XY at 10 s in 48/288 and 40/288 conditions. On ISAS the counts are 0/96 and 15/96. Pfeiffer's MHE accepts a position seed only and reaches the same threshold

TABLE V

Published-method endpoint comparison on Delft. Each entry is an RMSE in metres over 66 windows per layout. Constant and raw rows are separate shared-input comparisons. The single Profile-1 row uses the body model.

		Eight anchors			Four lowest anchors		
Estimator and configuration	Radio	XY	Z	3D	XY	Z	3D
Pacholska *et al.*, degree one, unweighted	Constant	.669	.425	.793	.689	5.803	5.844
Pacholska *et al.*, degree one, author weights	Constant	.672	.411	.788	.676	6.268	6.305
Range-only degree two, author weights	Constant	.356	.491	.606	.474	7.982	7.996
Pfeiffer *et al.*, author MHE	Constant	.121	.321	.343	.187	.876	.895
Pfeiffer *et al.*, author EKF	Constant	.105	.231	.254	.188	.869	.889
Single Profile-1 with body bias	Constant	.133	.245	.279	.204	.854	.878
Pfeiffer *et al.*, author MHE	Raw	.150	.549	.569	.219	.235	.321
Pfeiffer *et al.*, author EKF	Raw	.172	.506	.534	.229	.243	.334
Single Profile-1 with body bias	Raw	.205	.523	.562	.397	.241	.465

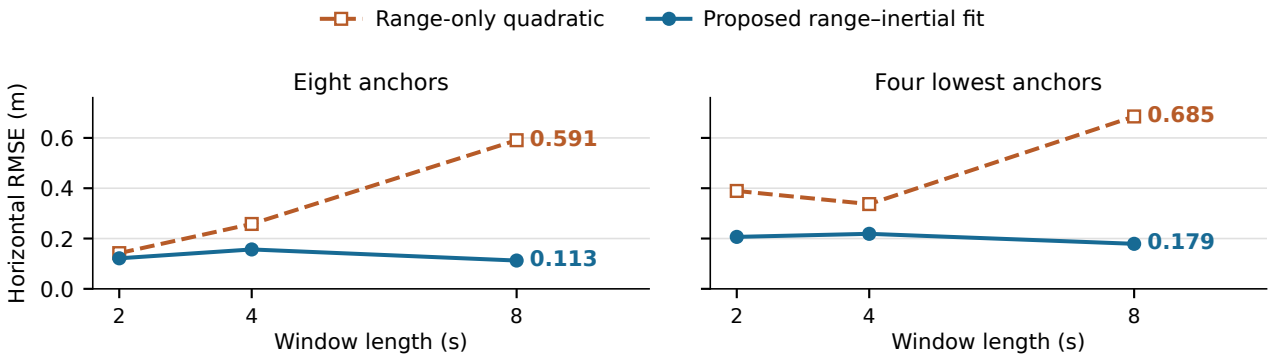


Fig. 3. Effect of measured inertial shape on horizontal endpoint RMSE. The quadratic range-only reconstruction and proposed fit share the range packets, constant calibration and endpoints. Each layout supplies 24, 24 and 18 windows at 2, 4 and 8 s. Both panels use the same linear scale. Table V includes the vehicle-model EKF and MHE.

TABLE VI

Cold start on the same packets. Time is median observed confirmation from the first IMU sample. XY is median horizontal error 10 s after prefix release. Held gives recordings within threshold from confirmation through 15 s.

	Delft, six flights			ISAS, two walks		
Estimator	Time (s)	XY (m)	Held	Time (s)	XY (m)	Held
Profile-1, body bias	4.5	.158	3/6	4.5	.346	0/2
SFUISE [3]	4.0	.095	6/6	3.0	.060	2/2
Pfeiffer EKF [2]	4.0	.084	5/6	Quadrotor model		
Pfeiffer MHE [2]	4.0	.094	6/6	Quadrotor model		
Jiang graph [4]	12.5	.075	1/6	12.5	.052	2/2

in all 96 distinct Delft position-seed trials. Each set of perturbations reuses one physical record per restart. This test concerns a supplied state error. The checkpoint experiment below varies actual ranging outage duration.

VII. Application to a checkpointed invariant filter

A. One checkpoint and one use of each range

The TWR filter propagates $x = (R, v, p, b_g, b_a)$ with a full 15-dimensional right-invariant-error covariance. To account for ranges already seen by the filter, we recompute one interval posterior from its pre-outage checkpoint and linearize about the acquired motion.

After a gap longer than 1 s, the live filter continues to emit estimates. We attempt one handoff after 2 s of new data if at least 16 positive ranges from four distinct anchors have arrived. The checkpoint c is the last complete pre-outage posterior or, at startup, the initial prior. The seed uses post-return ranges. The conditional interval $c{:}k$ retains IMU samples through the outage and every unique range received after c.

Let P_c denote the checkpoint covariance in the appropriate local chart, u_j the IMU sample and S_w the covariance of the native discrete process noise. In a local state difference $\boxminus$, the interval problem has the form

$$\begin{aligned}\min_{x_{c:k},w_{c:k-1}} \quad & \tfrac{1}{2}\,\|x_c \boxminus \hat{x}_c\|^2_{P_c^{-1}} + \tfrac{1}{2}\sum_{j=c}^{k-1}\|w_j\|^2_{S_w^{-1}} \\ & + \sum_{i\in\mathcal{E}_{c:k}} \rho_2\!\left(\frac{\|p_i - a_i\| + \beta_{a_i} - z_i}{\sigma_i}\right), \qquad (20)\\ & x_{j+1} = f_j(x_j, u_j, w_j).\end{aligned}$$

Here $\|z\|^2_M = z^{\mathsf{T}} M z$ and $\mathcal{E}_{c:k}$ excludes packets already in the checkpoint. The smoother seed uses a profiled affine fit followed by body-bias refinement. Its endpoint and bias penalty enter only through the initial trajectory. Each of four Kalman smoothing passes uses the original prior,

TABLE VII

Horizontal tracking 6–11 s after ranging resumes. Entries are medians of per-flight XY RMSE (m). The time series in Fig. 4 illustrates the 20 s indoor outage on one flight.

	Delft (6 flights)			QDrone (8 recordings)		
Gap (s)	Continue	Handoff	Lifted	Continue	Handoff	Lifted
2	.094	.093	.094	.551	.551	.551
5	.093	.093	.093	.545	.545	.545
10	.095	.092	.092	.635	.547	.544
20	39.429	.092	.092	.577	.533	.536

native inertial dynamics, bias random walks and the same packets, with range weights recomputed at the current trajectory. The continuing filter retains its innovation gate.

At handoff, we discard the provisional terminal posterior for the interval and take the full smoothed state and covariance. Denote the smoother's local error by $\delta x_P = (\delta\theta_b, \delta v_W, \delta p_W, \delta b_g, \delta b_a)$. After the final attitude retraction its covariance is reset by $G = \mathrm{diag}(J_r(\delta\theta_b), I_{12})$ with the right Jacobian J_r of $SO(3)$. For left injection into the right-invariant filter the coordinate map at the new endpoint is

$$A(X) = \begin{bmatrix} R & 0 & 0 & 0 \\ [v]_\times R & I & 0 & 0 \\ [p]_\times R & 0 & I & 0 \\ 0 & 0 & 0 & I_6 \end{bmatrix}, \qquad P_I = A\,G P_{\mathrm{sm}} G^{\mathsf{T}} A^{\mathsf{T}}. \tag{21}$$

This transports both navigation–bias and attitude–translation cross terms. For example, a body attitude perturbation becomes $R\delta\theta_b$ in world coordinates and contributes $[p]_\times R\delta\theta_b$ to the invariant translation error. The continuing filter therefore inherits the position–attitude coupling at the recovered state.

B. Same-history intervention on two TWR datasets

The IMU remains active during range outages of 2, 5, 10 or 20 s ending at $t = 25$ s. Two policies share the same startup and prehistory. H continues the InEKF, and W applies the window handoff. Their within-flight difference isolates recovery. The third policy L uses an earlier multi-anchor lifted position/velocity capture followed by an InEKF and supplies a benchmark with a different history.

The intervention produces 64 flight/gap pairs on 16 recordings. H and W agree in state, biases and covariance diagonals through the acquisition wait on every pair. Six indoor and eight outdoor recordings per gap meet the 95% reference-coverage criterion. Four outdoor attempts retain the native filter owing to insufficient anchor support.

On the triangle flight, the continued and handoff filters follow the same drifting state until the window is ready. The handoff then returns the estimate to the measured path (Fig. 4). Over $[31, 36)$ s, the handoff and lifted-capture filters both track it to 0.075 m horizontal RMSE. This score uses the latest emitted state at each scoring instant. The window fits in Fig. 2 contribute one position error per endpoint.

Across the six flights, a handoff after the 20 s outage reduces every late error (Table VII). The median of the paired changes is -39.352 m. In the 2–5 s interval the cohort median falls from 16.451 to 0.109 m. By 6–11 s it reaches the 0.092 m median of lifted capture. Continued filtering remains competitive after short indoor gaps.

The outdoor trial tests the same intervention across sites. After a 20 s gap the two late cohort medians are 0.577 and 0.533 m, with a median within-recording change of $+0.00056$ m. At 2 s, continuing the filter gives the lower late error on six of eight supported recordings. These paired results favor using the outdoor window first as a position fix.

C. Physical heading with the original TDoA InEKF

For the original filter of Oursland and Mehrabian [21], the published antenna offset ℓ enters the ideal signed observation for anchors a, b as

$$h_{ab}(R, p) = \|p + R\ell - b\| - \|p + R\ell - a\|\,. \tag{22}$$

This observation can change instantaneously with orientation through the lever arm. In the zero-lever-arm TWR model (2), heading affects distances through the trajectory history. We therefore assess physical yaw with the original InEKF class on Toronto UTIL flights with motion-capture attitude [24]. Each cohort comprises four recordings and two initial-yaw interventions (0° and 180°). Both arms use a common sensor-only initializer on the first 0.6 s, then retain the original event queue and covariance. We suppress TDoA for $[10, 20)$ s and score wrapped yaw and XY on $[26, 45)$ s.

Reserving packets for an early XY update changes median horizontal/yaw RMSE from 0.112 m/5.93° to 0.087 m/6.80° in the primary cohort, and from 0.0705 m/7.22° to 0.0709 m/6.22° in the complementary cohort. Position and heading can thus improve in different cohorts.

A separate cold-start replay uses trial four in each Toronto layout, the same TDoA and IMU packets, and the same antenna offset for both methods. With a 1.5 s sensor-only constant-velocity initialization, the Oursland filter confirms at a median 3.0 s from the first raw sample. SFUISE confirms at 1.5 s. At 15 s the median XY/heading errors are 0.049 m/4.21° for Oursland and 0.082 m/7.22° for SFUISE. Four and three recordings, respectively, remain inside the horizontal threshold after confirmation. The absolute-range initializer in Table VI was evaluated on different measurements.

VIII. Discussion and conclusion

For a window with unknown heading, polynomial elimination reduces candidate generation to minimization on a unit circle. In the body-bias fits, one such seed attains the same tested objective as four uniform starts across

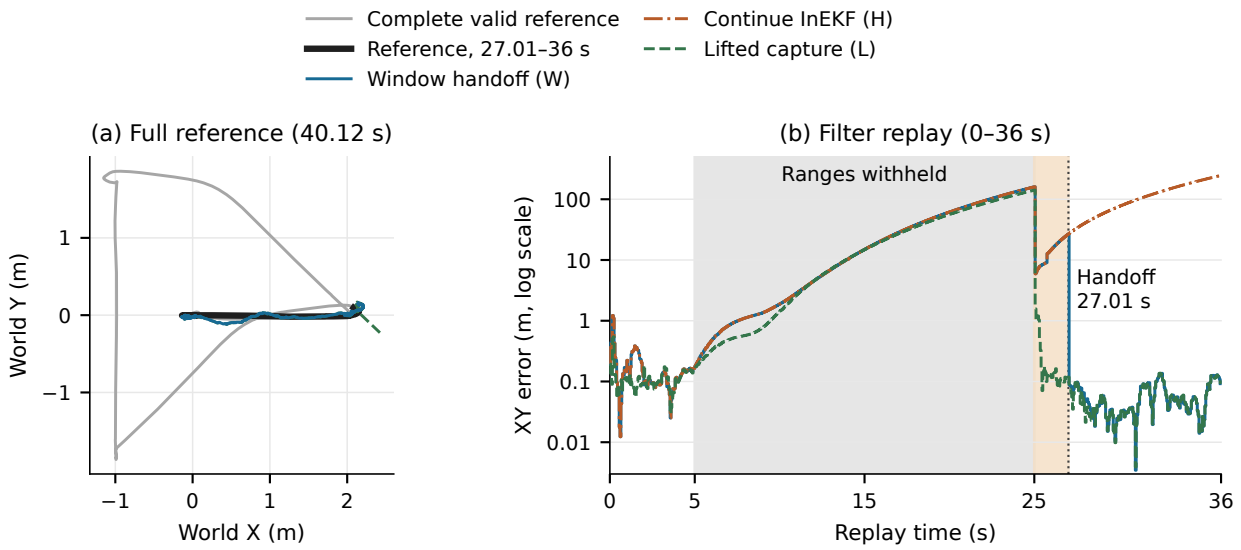


Fig. 4. Delft triangle, first in the fixed six-flight protocol. Left, the full valid motion-capture path spans 40.12 s. The highlighted reference and continuous handoff and lifted-capture estimates cover 27.01–36 s on equal spatial axes. Right, horizontal error for all three policies using the latest state at 20 Hz over $[0, 36)$ s. Ranges were withheld over $[5, 25)$ s. The first return arrived at 25.01 s and handoff followed at 27.01 s.

all 66 Delft windows and their layout and coordinate variants. The fitted objective and the physical endpoint measure different properties. The local-start control in Table IV has slightly lower pooled XY error despite seven objective misses. The savings arise when competing heading hypotheses would otherwise require additional distance fits.

For the correctly specified noiseless lift, the projected rank condition is sufficient for heading uniqueness. With a fixed design, (10) also depends on the size of the projected disturbance. The singular value alone cannot establish physical heading accuracy. The conditional curvature in (17) measures local range sensitivity after refitting nuisance states. A free world acceleration and a body-fixed bias agree to second order under smooth rotation. Their pooled Delft horizontal errors differ by only 0.004 m under directional calibration. A moving IMU prefix adds a further limitation. In one ISAS restart, averaging the prefix produced a nominal gyro bias of 0.165 rad/s in norm while the independent reference showed mean angular velocity of 0.157 rad/s in norm. The resulting time-varying orientation error is outside the constant accelerometer-offset model.

With a pre-outage filter checkpoint, relinearizing the return interval matched the older lifted capture's median late XY error after the 20 s Delft gap. The outdoor median paired change was approximately zero. For unknown-position startup, SFUISE confirmed a usable fix sooner and maintained it on more recordings. The evidence supports a short-window position initializer with a directly testable heading search. Using its heading to reset a running filter would require independent validation across motion and anchor geometries.

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